Landslide Early Warning System

MATLAB–Simulink Based AI Sensor Fusion and Time-to-Failure Prediction

Group Name: Palliyanagdi Musclepidithakaar

Members

Arosh Rolly

Ammar Mohammad

Ramees A.K

Sanjay P.R

Abstract

Landslides in the Western Ghats — particularly in Kerala's Munnar–Idukki region — are strongly influenced by prolonged, intense rainfall. This project presents a MATLAB/Simulink-based Landslide Early Warning System (LEWS) that monitors the progression of rainfall-induced slope instability and issues multi-level warnings in real time. The system models the physical cascade from rainfall infiltration to rising soil moisture, increasing pore water pressure, declining Factor of Safety (FS), and growing surface deformation and subsurface strain.

Six sensor inputs — rainfall, volumetric soil moisture, pore water pressure, FS, surface tilt, and shear strain — feed an AI-inspired sensor-fusion stage that computes a composite risk percentage and classifies slope condition into four warning levels: Green, Yellow, Orange, and Red. A Time-to-Failure (TTF) prediction stage further estimates the remaining time before failure based on evolving sensor trends. The Simulink model integrates physical-process blocks, AI-based fusion logic, warning-decision logic, and a single multi-state visual indicator for real-time-style demonstration.

The system is validated using a Munnar/Idukki rainfall-event sequence, with simulation time compressed such that one simulated second represents 0.6 hours of physical time. Results show the expected progression from stable conditions to increasing deformation, decreasing FS, falling TTF, and eventual failure — demonstrating a proof-of-concept framework for combining conventional geotechnical indicators with AI-based sensor fusion in landslide early warning.

Keywords

Landslide Early Warning System; Munnar; Idukki; Western Ghats; MATLAB; Simulink; AI Sensor Fusion; Factor of Safety; Pore Water Pressure; Surface Tilt; Shear Strain; Risk Assessment; Time-to-Failure.

Project Focus

This project's core contribution is a unified monitoring framework that fuses hydrological, geotechnical, and deformation-related parameters into a single risk assessment, rather than relying on rainfall thresholds alone — an approach commonly used in conventional systems but prone to false alarms and missed events on complex slopes. By combining rainfall, moisture, pore pressure, FS, tilt, and strain data through AI-based sensor fusion, the system captures the interdependent physical processes that precede slope failure, enabling earlier and more reliable detection of instability.

The framework's dual output — a staged warning classification alongside a continuously updated Time-to-Failure estimate — is designed to give decision-makers both an immediate risk indicator and a forward-looking planning horizon, supporting timely evacuation and response decisions in vulnerable Western Ghats communities.

Note

> The current implementation is a simulation and proof-of-concept model. AI estimates are based on the available project dataset and should not be interpreted as field-validated probabilities or operational predictions without larger, independently labelled, site-calibrated datasets.